Theorem Certificate

A new, small, checked lemma with your name on it

Lemma named for

RESERVED

Serial: bat-1e5dfc2e007e

Lean 4 source

```
import Mathlib
-- Buyer: RESERVED
-- Serial: bat-1e5dfc2e007e
theorem bat_1e5dfc2e007e : 291 ^ 2 + 326 ^ 2 >= 2 * 291 * 326 := by
norm_num
```

In plain English:

The sum of the squares of 291 and 326 is at least twice their product.

Pin

Lean 4.35.0, Mathlib commit 56a005855ea1 (full hash below)

Mathlib commit (full hash)

56a005855ea17e0b0958f73ec801ae140b0a1679

Proof (Lean file) sha256

d4571003f86fc328a1ece01d282bdbdd19558ab7b1fa7e308fb5825288c7a90b

Check log sha256

e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855 (empty log: Lean printed no messages)

Verify (signature only)

```
python tools/attest_verify.py bat-1e5dfc2e007e.dsse.json --key <pubkey> --expect-keyid <keyid>
```

Verify (full)

python -m buyatheorem verify bat-1e5dfc2e007e (needs a local Lean 4 + Mathlib project; see README)

New means: statement text not found verbatim in Mathlib and not previously minted. No mathematical importance is claimed.

Precedent: TheoryMine (2010).

The buyer or recipient name is PUBLIC in the registry, the Lean file and this PDF.